6 Algebra, equations and inequalities

Sheet 31 Thu 14 Jan. 2027 *Integer powers (positive and negative)*

1. Warm-up · Quick [PUI] Calculate 2^{-3}, 10^{-2} and $(-3)^2$.
   Ans. $\frac{1}{8}$; 0.01; 9.
2. Core · Method [PUI] Simplify $\dfrac{2^3 \times 2^{-5}}{2^{-1}}$.
   Ans. $\dfrac{2^{-2}}{2^{-1}} = 2^{-2+1} = 2^{-1} = \dfrac{1}{2}$.
3. Core + · Method [PUI] For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.
   Ans. $x^{-6} \times x^7 = x^1 = x$.
4. Review · Quick [STA] Median of the series 4, 7, 9, 10, 12, 15?
   Ans. Even number of values: $\dfrac{9+10}{2} = 9.5$.
5. Thread · Identify [SOL] Which solid has a disc-shaped base, a curved lateral surface and a single vertex?
   Ans. A right circular cone.

Sheet 32 Mon 18 Jan. 2027 *Square roots; $\sqrt{a^2} = |a|$*

1. Warm-up · Quick [CAL] Calculate $\sqrt{36}$ and $\sqrt{0.09}$.
   Ans. 6; 0.3.
2. Core · Method [CAL] Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.
   Ans. $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$; $\sqrt{(-5)^2} = |-5| = 5$.
3. Core + · True/false [CAL] True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.
   Ans. False. $\sqrt{25} = 5$ but $3 + 4 = 7$. Only the product "splits": $\sqrt{ab} = \sqrt{a}\sqrt{b}$.
4. Review · Quick [PUI] Calculate $(3^2)^3$.
   Ans. $3^{2\times 3} = 3^6 = 729$.
5. Thread · True/false [DEM] True or false? Justify. The diagonals of a rhombus are perpendicular.
   Ans. True. Property of the rhombus (its diagonals are also perpendicular bisectors of each other).

Sheet 33 Tue 19 Jan. 2027 *Fractional expressions; expanded, reduced or factorised forms*

1. Warm-up · Quick [CAL] Calculate $\dfrac{1}{2} + \dfrac{1}{3}$.
   Ans. $\dfrac{5}{6}$.
2. Core · Method [LIT] Write as a single fraction $\dfrac{1}{x} + \dfrac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).
   Ans. $\dfrac{x+1+x}{x(x+1)} = \dfrac{2x+1}{x(x+1)}$.
3. Core + · Method [DEV] Factorise $x^2 - 4$ then simplify $\dfrac{x^2-4}{x-2}$ for $x \neq 2$.
   Ans. $x^2 - 4 = (x-2)(x+2)$; so $\dfrac{x^2-4}{x-2} = x + 2$.
   The choice of form depends on the goal: factorised to simplify, expanded to compare.
4. Review · Quick [SUB] Calculate $2a - 3b$ for $a = 2$ and $b = -1$.
   Ans. $4 + 3 = 7$.
5. Thread · Quick [REP] $ABCDEFGH$ is a cuboid with $AB = 4$, $AD = 3$, $AE = 2$. In the frame

(A, B, D, E), what are the coordinates of G?

Ans. $G(4,3,2)$: G is the vertex opposite A.

A third coordinate appears in space.

Sheet 34 Thu 21 Jan. 2027 *Inequalities: sum, multiplication by a real number*

1. Warm-up · Quick [CMP] Complete with $<$ or $>$: $-3 \ldots -2$; $\frac{1}{3} \ldots 0.3$.

   Ans. $-3 < -2$; $\frac{1}{3} > 0.3$ ($0.333\ldots > 0.3$).

2. Core · Method [EQ] We have $a < b$. Compare $-2a$ and $-2b$, then $a+5$ and $b+5$.

   Ans. $-2a > -2b$ (multiplying by a negative number reverses the inequality); $a+5 < b+5$.

3. Core + · Method [EQ] Given $2 < x < 3$ and $1 < y < 4$, find bounds for $x+y$ then $x-y$.

   Ans. $3 < x+y < 7$; for $x-y$: $-y \in (-4,-1)$ so $-2 < x-y < 2$.

   Pitfall: subtracting the inequalities term by term.

4. Review · Quick [CAL] Calculate $(-3) \times (-4) + 2 \times (-5)$.

   Ans. $12 - 10 = 2$.

5. Thread · Problem [DIV] A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side length?

   Ans. $\gcd(120, 84) = 12$: squares of side 12 cm ($10 \times 7 = 70$ squares).

Sheet 35 Mon 25 Jan. 2027 *Comparing by difference and by quotient*

1. Warm-up · Quick [CMP] Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of the difference.

   Ans. $\frac{2}{3} - \frac{3}{4} = -\frac{1}{12} < 0$: $\frac{2}{3} < \frac{3}{4}$.

2. Core · Method [LIT] Compare x^2+1 and $2x$ for every real number x.

   Ans. $x^2 + 1 - 2x = (x-1)^2 \geqslant 0$: $x^2 + 1 \geqslant 2x$, with equality if and only if $x = 1$.

3. Core + · Method [CMP] Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).

   Ans. $\frac{7/5}{8/6} = \frac{42}{40} > 1$, so $\frac{7}{5} > \frac{8}{6}$.

4. Review · Table [LEC] Monthly rainfall (fictional data).

Month	Jan.	Feb.	Mar.	Apr.
Rain (mm)	280	300	240	120

   Which month is the wettest? What is the total over these 4 months?

   Ans. February (300 mm); 940 mm.

5. Thread · Table → formula [REG] x: -2, -1, 0, 1 and $f(x)$: 4, 1, 0, 1. Is the function affine? Suggest a formula.

   Ans. No (the changes are not constant). $f(x) = x^2$ works.

   Test: $f(1) - f(0) = 1$ but $f(0) - f(-1) = -1$.

Sheet 36 Tue 26 Jan. 2027 *First-degree equations and inequalities; $x^2 = a$; zero product*

1. Warm-up · Quick [EQ] Solve $3x - 5 = 7$.
   Ans. $x = 4$.
2. Core · Method [EQ] Solve $2x + 3 \leqslant 7 - x$.
   Ans. $3x \leqslant 4$: $x \leqslant \frac{4}{3}$; $S = \left]-\infty; \frac{4}{3}\right]$.
3. Core + · Method [EQ] Solve $(x - 2)(2x + 6) = 0$ then $x^2 = 7$.
   Ans. Zero product: $x = 2$ or $x = -3$; $x^2 = 7$: $x = \sqrt{7}$ or $x = -\sqrt{7}$.
4. Review · Quick [CMP] Compare 1.4 and $\sqrt{2}$ without a calculator.
   Ans. $1.4^2 = 1.96 < 2$ so $1.4 < \sqrt{2}$ (the square function is increasing on $[0, +\infty)$).
5. Thread · Quick [GEO] ABC is isosceles at A with $\widehat{BAC} = 40°$. What is the measure of $\widehat{ABC}$?
   Ans. $(180 - 40) \div 2 = 70°$.

Sheet 37 Thu 28 Jan. 2027 *Quotient equation $\frac{A(x)}{B(x)} = k$*

1. Warm-up · Quick [EQ] For which value of x is the expression $\dfrac{1}{x - 3}$ undefined?
   Ans. $x = 3$.
2. Core · Method [EQ] Solve $\dfrac{2x - 1}{x + 3} = 0$.
   Ans. $x \neq -3$ and $2x - 1 = 0$: $x = \frac{1}{2}$.
3. Core + · Method [EQ] Solve $\dfrac{x + 1}{x - 2} = 3$.
   Ans. $x \neq 2$; $x + 1 = 3(x - 2)$, that is $x = \frac{7}{2}$. Check: $\frac{4.5}{1.5} = 3$.
4. Review · Expand [DEV] Expand and simplify $(x + 2)(x - 5)$.
   Ans. $x^2 - 3x - 10$.
5. Thread · Code reading [ALG] What do $g(3)$ and $g(-5)$ return?

```
def g(x):
    if x > 0:
        return 2*x
    else:
        return -x
```

   Ans. $g(3) = 6$ and $g(-5) = 5$.
   The condition is evaluated before the calculation: the branch depends on the sign of x.

Sheet 38 Mon 1st Feb. 2027 *Sign of a product or quotient; sign tables*

1. Warm-up · Quick [EQ] Sign of $2x - 4$?
   Ans. Negative for $x < 2$, zero at 2, positive for $x > 2$.
2. Core · Method [EQ] Draw the sign table of $(x - 1)(x + 3)$.
   Ans. Roots -3 and 1: positive on $(-\infty, -3)$, negative on $(-3, 1)$, positive on $(1, +\infty)$.
   Sign of the product: rule of signs, column by column.
3. Core + · Method [EQ] Solve $\dfrac{x + 2}{x - 1} \geqslant 0$.
   Ans. Numerator positive for $x \geqslant -2$, denominator positive for $x > 1$; $S = (-\infty, -2] \cup (1, +\infty)$ (1 is excluded).
4. Review · Quick [PRB] Pack of 32 cards: probability of drawing a heart? a king?

Ans. $\frac{8}{32} = \frac{1}{4}$; $\frac{4}{32} = \frac{1}{8}$.

5. Thread · Disprove [LIT] Disprove: "for every real number x, $x^2 \geqslant x$".

Ans. Counterexample: $x = \frac{1}{2}$: $x^2 = \frac{1}{4} < \frac{1}{2} = x$.

A single counterexample is enough to disprove a universal statement.